

5.

Ans: D

Let L be the distance from hip joint pivot to centre of mass.

Applying principle of moment at the hip joint pivot,

$$TL \sin 8^\circ = WL \sin 70^\circ$$

$$T = W \frac{\sin 70^\circ}{\sin 8^\circ}$$

$$= 400 \times 6.752$$

$$= 2700 \text{ N}$$